ALLEN

Minor Test-1 (Phase – 11&14)

CLASS-XII PASS

NEET LEADER

TIME : 180 Min **Maximum Marks : 720**
Date : 09.09.2026

CODE: 4504CMD303039261001

Instructions

1. When you are directed to attempt the Test Paper, **fill in the particulars in the given OMR Sheet** carefully with **blue/black** ball point pen only.
2. The test is of **180 minutes** duration and the Test Booklet contains **180** multiple-choice questions (four options with a single correct answer).
3. This paper is divided into **3 Parts** consisting of **Physics, Chemistry, Biology (Botany and Zoology).**
4. The paper will consist of **45** questions each in Physics, Chemistry and **90** questions in Biology.
5. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each **incorrect** response, **one mark will be deducted** from the total score. **The maximum marks are 720.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on OMR Sheet.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc., except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the OMR Sheet to the invigilator on duty in the Room/Hall.
10. Do not fold or make any stray marks on the OMR Sheet.
11. Students are instructed to Mark Answer / Bubble Answers in from Question No. 1 to 180 as per Question Paper Series only in OMR

    Important :- In OMR Question No. 181 to 200 has to be left blank.

Name of the Candidate : .....

Form Number : .....

SUBJECT : PHYSICS

1. Given that $y = \dfrac{10}{\sin x + \sqrt{3}\cos x}$. Minimum value of y is.

(1) zero (2) 2

(3) 5 (4) $10/(1+\sqrt{3})$

2. A particle moves along the curve $12y = x^3$. Which coordinate changes at faster rate at x = 10?

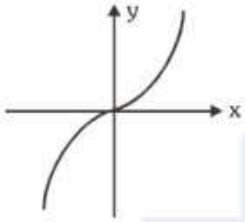
(1) x-coordinate

(2) y-coordinate

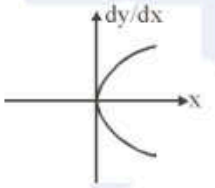
(3) Both x and y-coordinate

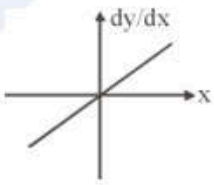
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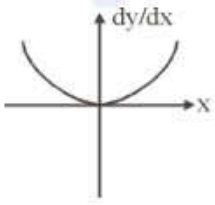
3. Figure shows the graph of the curve $y = \dfrac{1}{3}x^3$.

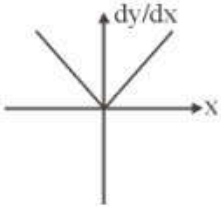


Which of the following shows the variation of slope of the curve with x coordinate ?

(1) 

(2) 

(3) 

(4) 

4. Double differentiation of $y = \ell n\, x^2 + \sin x$ w.r.t. x is :

(1) $\dfrac{2}{x} + \cos x$ (2) $\dfrac{-2}{x^2} + \cos x$

(3) $\dfrac{-2}{x^2} - \sin x$ (4) $\dfrac{+2}{x} - \sin x$

5. Given that $y = \sin 3x + \dfrac{4}{3}\cos 3x$. What is the maximum rate of change in y with respect to x ?

(1) 7 (2) 4 (3) 5 (4) 6

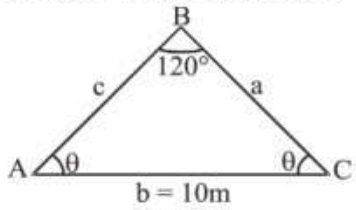
6. Electric current in a wire is time rate of flow of charge. The charge in coulombs that passes through a wire after t seconds is given by the function $q(t) = t^3 - 2t^2 + 5t + 2$. Determine the average current (in coulomb per second) during the first two seconds. $\left[< I >= \dfrac{\Delta q}{\Delta t}\right]$

(1) 4A (2) 5A (3) 6A (4) 12A

7. Value of tan(106°) will be :-

(1) $\dfrac{24}{7}$ (2) $-\dfrac{7}{24}$ (3) $-\dfrac{24}{7}$ (4) $-\dfrac{8}{3}$

8. For a triangle shown in the figure, side CA is 10 m, angle ∠A and ∠C are equal then :



(1) a = c = 10m (2) $a = c = \dfrac{10}{\sqrt{3}}$m

(3) $a \neq c$ (4) $a = c = \dfrac{10}{\sqrt{2}}$m

9. Value of tan 37° is

(1) 1 (2) 3/5

(3) 4/5 (4) 3/4

10. Which of the following graph is/are straight line for the equation $y^2 = 2x$?

I. Graph: y versus x^2

II. Graph: y^2 versus x

III. Graph: y versus $\sqrt{x}$

IV. Graph: $\sqrt{y}$ versus x

(1) I, IV (2) II, III

(3) I, III (4) Only II

11. If the volume of a sphere increases at constant rate $\left(\frac{dV}{dt} = 4\right)$. If radius of the sphere is denoted by r, then surface area of the sphere increases at the rate :-

(1) $\frac{4}{r}$ (2) $\frac{8}{r}$

(3) $\frac{12}{r}$ (4) $\frac{16}{r}$

12. Find the equation of the line passing through (2, −1) and parallel to the line $2x - Y = 4$:-

(1) $Y = 3x - 4$ (2) $Y = 2x + 4$

(3) $Y = 5x - 2$ (4) $Y = 2x - 5$

13. If energy(E), area(A) and velocity(V) are taken as base quantities, then dimensional formula for planck's constant is :-

(1) $E A^{1/2} v^{-1/2}$ (2) $E A^{1/2} v^{-1}$

(3) $E^2 Av^{-1}$ (4) $E A^2 v^{-2}$

14. In a vernier callipers, N division of Vernier scale coinside with (N−1) division of main scale (in which 1 division represent 1 mm). The least count of the instrument in cm should be :-

(1) N (2) N−1

(3) $\frac{1}{10N}$ (4) $\frac{1}{N-1}$

15. Two full turns of the circular scale of gauge cover a diastance of 1 mm on scale. The total number of divisions on circular scale is 50. Further, it is found that screw gauge has a zero error of −0.03 mm. While measuring the diameter of a thin wire a student notes the main scale reading of 3 mm and the number of circular scale division in line, with the main scale as 35. The diameter of the wire is

(1) 3.32 mm (2) 3.73 mm

(3) 3.67 mm (4) 3.38 mm

16. The resistance R of a wire is given by the relation $R = \frac{\rho \ell}{\pi r^2}$. Percentage error in the measurment of ρ, ℓ and r is 1%, 2% and 3% respectively. Then the percentage error in the measurment of R is :−

(1) 6% (2) 9% (3) 8% (4) 10%

17. Dimensions of an unknown quantity $\phi = \frac{ma}{\alpha}\log\left(1 + \frac{\alpha \ell}{ma}\right)$, . Where m = mass, a = acceleration and ℓ = length are :-

(1) $[MLT^2]$ (2) $[MT^{-2}]$

(3) $[M^0LT^0]$ (4) $[MT^{-3}]$

18. Find the dimensions of inductance :-

(1) $[ML^2T^{-2}A^{-2}]$ (2) $[ML^2T^{-1}A^{-1}]$

(3) $[MLT^{-1}A^{-1}]$ (4) $[MLTA^{-2}]$

19. If force (F), work(W) and velocity (V) are taken as fundamental quantities, then the dimensional formula of time (t) is :-

(1) [WFV] (2) $[WFV^{-1}]$

(3) $[W^{-1}F^{-1}V^{-1}]$ (4) $[WF^{-1}V^{-1}]$

20. The weight of an object in air is (5.00 ± 0.05) N and weight of the object in water is (4.00 ± 0.05)N. Find the maximum percentage error in measurement of difference in weights:

(1) 1% (2) 5%

(3) 10% (4) 2%

21. Force F and density d are related as $F = \frac{\alpha}{\beta + \sqrt{d}}$ then the dimensions of α and β are :-

(1) $M^{3/2}L^{-1/2}T^{-2}$, $M^{1/2}L^{-3/2}$ respectively

(2) $M^{-3/2}L^{1/2}T^{-2}$, $M^{-1/2}L^{3/2}$ respectively

(3) $M^{3/2}L^{-1/2}T2$, $M^{-1/2}L^{3/2}$ respectively

(4) $M^{3/2}L^{1/2}T^{-2}$, $M^{1/2}L^{3/2}$ respectively

22. The number 2.745 and 2.735 on rounding off to 3 significant figures will give :-

(1) 2.75 and 2.74 (2) 2.74 and 2.73

(3) 2.75 and 2.73 (4) 2.74 and 2.74

23. A body travels uniformly a distance of (13.8 ± 0.2) m in a time (4.0 ± 0.3) s. Its velocity with error limits is :-

(1) $(3.4 \pm 0.6)\ ms^{-1}$ (2) $(3.4 \pm 0.4)\ ms^{-1}$

(3) $(6.1 \pm 0.6)\ ms^{-1}$ (4) $(6.1 \pm 0.3)\ ms^{-1}$

24. $\alpha = \frac{F}{v^2}\sin(\beta t)$ (here v = velocity, F = force, t = time) Find dimension of α & β :-

(1) $\alpha = [M^1L^1T^0]$, $\beta = [T^{-1}]$

(2) $\alpha = [M^1L^1T^{-1}]$, $\beta = [T^1]$

(3) $\alpha = [M^1L^1T^{-1}]$, $\beta = [T^{-1}]$

(4) $\alpha = [M^1L^{-1}T^0]$, $\beta = [T^{-1}]$

25. In a particular system the units of length mass and time are chosen to be 10 cm, 10 g and 0.1 s respectively. The unit of force in this system will be equal to

(1) 0.1 N (2) 1 N

(3) 10 N (4) 100 N

26. A force F is given by $F = at + bt^2$, where t is time. The dimensions of a and b : ––

(1) $[MLT^{-3}]$ and $[MLT^{-4}]$

(2) $[MLT^{-4}]$ and $[MLT^{-3}]$

(3) $[MLT^{-1}]$ and $[MLT^{-2}]$

(4) $[MLT^{-2}]$ and $[MLT^0]$

27. Potential energy of a particle moving along x-axis is given as $U = (x^3 + 3x^2 - 9x + 6)$ J. Then find minimum possible value of U

(1) –21J (2) –4J

(3) 1J (4) –3J

28. If position of particle varies with time as $S = 2t^2 + t + 1$ m, then its speed (velocity) at t = 1 sec will be $\left[V = \frac{ds}{dt}\right]$:-

(1) 5 m/s (2) 6 m/s

(3) 2 m/s (4) 1 m/s

29. If $\cos\theta = \frac{7}{25}$ and 'θ' lies in 1st quadrant, then find $\tan(90° - \theta)$:-

(1) $\frac{24}{7}$ (2) $\frac{24}{25}$ (3) $\frac{7}{24}$ (4) $\frac{25}{24}$

30. Convert 2160" into degree :

(1) 36° (2) 3.6°

(3) 0.6° (4) 0.36°

31. If $3\cos\theta + 4\sin\theta = A\sin(\theta + \alpha)$, then values of A and α are

(1) $5,\ \cos^{-1}\left(\frac{3}{5}\right)$ (2) $5,\ \sin^{-1}\left(\frac{3}{5}\right)$

(3) $7,\ \sin^{-1}\left(\frac{4}{5}\right)$ (4) None of these

32. The velocity of a particle moving along x-axis is given as $V = 4t - t^2$. The maximum and minimum velocity of particle in the interval $1 \le t \le 4$ is :-

(1) $4ms^{-1}, 3ms^{-1}$ (2) $4ms^{-1}, 1ms^{-1}$

(3) $4ms^{-1}, 0ms^{-1}$ (4) $0ms^{-1}, 4ms^{-1}$

33. $y = \log_e(\sin x), \frac{dy}{dx} = ?$

(1) $\tan x$ (2) $\cos x$

(3) $\frac{1}{\sin x}$ (4) $\cot x$

34. If $y = \frac{x}{\sin x}$ find $\frac{dy}{dx}$:-

(1) $\frac{\sin x + x\cos x}{\sin^2 x}$ (2) $\frac{\sin x - x\cos x}{\sin^2 x}$

(3) $\frac{x}{\cos x}$ (4) $\frac{\sin x + x^2\cos^2 x}{\cos x}$

35. A stone is dropped into a quite lake and waves in circles spreading out radially at the speed of 0.25 m/s. At the instant when the radius of the circular wave is $\frac{4}{\pi}$ m, how fast is the enclosed area increasing?

(1) $\frac{4}{\pi}m^2/s$ (2) $\frac{2}{\pi}m^2/s$

(3) $2\ m^2/s$ (4) $\frac{1}{2}m^2/s$

36. Find the value of

$\frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \ldots\ldots\infty$

(1) $\frac{2}{3}$ (2) $\frac{1}{3}$

(3) 3 (4) $-\frac{2}{3}$

37. $\sqrt{2 + \sqrt{2 + 2\cos 4\theta}} = ?$

(1) $2\sin\theta$ (2) $2\cos\theta$

(3) $\sin\theta$ (4) $\cos\theta$

38. Given that $P = \frac{4}{\sqrt{4 + \frac{r^2}{2}}}$ and $(r << 2)$, then P is roughly equal to :

(1) $\left(1 - \frac{r^2}{16}\right)$ (2) $\left(2 - \frac{r^2}{8}\right)$

(3) $\left(1 - \frac{r}{16}\right)$ (4) $\left(2 - \frac{r}{8}\right)$

39. Then find rate of change of y if $y = \sin^2\sqrt{x}$:-

(1) $\frac{\cos\sqrt{x}}{2\sqrt{x}}$ (2) $\frac{\sin 2\sqrt{x}}{2\sqrt{x}}$

(3) $\frac{\sin\sqrt{x}}{\sqrt{x}}$ (4) $\frac{\sin\sqrt{x}}{2\sqrt{x}}$

40. Value of $\sin75°$ is

(1) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (2) $\cos15°$

(3) $\cos105°$ (4) All of the above

41. The equation of the stationary wave is :-

$y = 2A\sin\left(\frac{2\pi ct}{\lambda}\right)\cos\left(\frac{2\pi x}{\lambda}\right)$

Which of the following statements is wrong ?

(1) The unit of ct is same as that of λ

(2) The unit of x is same as that of λ

(3) The unit of $\left(\frac{2\pi c}{\lambda}\right)$ is same as that of $\left(\frac{2\pi x}{\lambda t}\right)$

(4) The unit of $\frac{c}{\lambda}$ is same as that of $\frac{x}{\lambda}$

42. E, m, L and G denote energy, mass, angular momentum and gravitational constant respectively; $\frac{EL^2}{m^5G^2}$ has the dimensions of

(1) length (2) mass

(3) time (4) angle

43. Which of the following have the same dimensions?

(1) Impulse and linear momentum

(2) Specific heat and latent heat

(3) Moment of inertia and force

(4) Thrust and surface tension

44. A travelling microscope has a main scale graduated in $\frac{1}{2}$ mm. 50 vernier divisions coincide with 49 main scale divisions. What will be the reading if the main scale reads 7.45 cm and the 29th division of the vernier coincides with the main scale division ?

(1) 7.459 cm (2) 7.453 cm

(3) 7.479 cm (4) 7.48 cm

45. If the units of force, energy and velocity in a new system be 20 N, 10 J and 10 m/s respectively, then the unit of distance in this system is :

(1) 0.25 m (2) 0.5 m

(3) 1 m (4) 2 m

46. The total number of atoms of oxygen in 0.5 mol of $Na_2B_4O_7.10H_2O$

(1) 8.5 N_A (2) 17 N_A

(3) 5 N_A (4) 10 N_A

47. The number of carbon atoms in 6.8g of C_3O_2 is :-

(1) 18.06×10^{23} (2) 18.06×10^{22}

(3) 6.023×10^{23} (4) 6.023×10^{22}

48. The weight of a molecule of the compound $C_6H_{12}O_6$ is about :

(1) 180 g (2) 3×10^{-22}g

(3) 22×10^{-23} g (4) 132 g

49. In which of the following pairs, do 1 g of each have an equal number of molecules ?

(1) N_2O and CO (2) N_2 and CO

(3) N_2O and CO_2 (4) Both 2 and 3

50. 3 mole of mixture of C_3H_8 and C_4H_{10} on complete combustion gives 10 mole CO_2 find mole % of C_3H_8 in the sample :-

(1) 33.34 (2) 66.66

(3) 30% (4) 50%

51. The density of water is 1 gm/ml what is the volume occupied by 1 molecule of water ?

(1) 1.44×10^{-23} ml (2) 1 ml

(3) 2.88×10^{-23} ml (4) 18 ml

52. 19.7 kg of gold was recovered from a smuggler. How many atoms of gold were recovered ? (At. wt. of gold (Au) = 197)

(1) 100 (2) 6.023×10^{23}

(3) 6.023×10^{24} (4) 6.023×10^{25}

53. Which of the following has maximum mass :-

(1) 0.1 gram atom of carbon

(2) 0.1 mol of ammonia

(3) 6.02×10^{22} molecules of hydrogen

(4) 1120 cc of carbon dioxide at STP

54. Equal mass of oxygen, hydrogen and methane are taken in a container in identical conditions. What is the ratio of their moles :-

(1) 1 : 16 : 1 (2) 1 : 16 : 2

(3) 8 : 1 : 8 (4) 16 : 1 : 8

55. At STP, 11.2 L of a gas weights 20g. The vapour density of gas is :-

(1) 32 (2) 40 (3) 20 (4) 8

56. Which of the following pairs have same number of molecules?

(a) 2 gm mole O_2, 4 gm mole SO_2

(b) 2 gm mole CO_2, 2 gm mole N_2O

(c) 224 ml O_2 at STP, 448 ml of He at 0.5 atm and 273 K

(d) 2 gms of oxygen and 2 gms of ozone

(1) a, b, c only (2) b, c only

(3) a, b, d only (4) a, b, c, d

57. When mixture of 6 L of CH_4 and Propane were burnt 7 L CO_2 was obtained then CH_4 and propane in mixture will be :-

(1) 5.5 L and 0.5 L (2) 6 L and 1 L

(3) 3 L and 4 L (4) 5 L and 2 L

58. Volume of air at STP containing 30% oxygen by volume, required to completely burn 200 L methane gas at STP would be

(1) 200 L (2) 400 L

(3) 1333.33 L (4) 2666.66 L

59. The percentage of Se in peroxidase enzyme is 0.5% by mass (atomic mass of Se = 78.4 amu). Then, the minimum molecular mass of enzyme :-

(1) 1.568×10^4 amu (2) 1.568×10^7 amu

(3) 1.568×10^3 amu (4) 1.568×10^6 amu

60. A metal M forms a metal carbonate M_2CO_3. If the mass % of oxygen in metal carbonate is 48% then the atomic weight of the metal is :-

(1) 40 (2) 20 (3) 60 (4) 80

61. In a compound C, H, N atoms are present in 9 : 1 : 3.5 ratio by mass. Molecular mass of compound is 108. Its molecular formula is :-

(1) $C_2H_6N_2$ (2) C_3H_4N

(3) $C_6H_8N_2$ (4) $C_8H_{12}N_2$

62. 12 moles of each A and B are allowed to react as given $3A + 2B \longrightarrow C + \frac{1}{2} D$. If 60 gm of D is produced then, calculate the atomic mass of D :-

(1) 30 (2) 45 (3) 60 (4) 15

63. What weight of 95% $CaCO_3$ will be required to produce 11.2 litre CO_2 :-

(1) 50 g (2) 100 g

(3) 52.6 g (4) 25 g

64. A metal oxide has 40% oxygen. The equivalent weight of the metal is :

(1) 12 (2) 16 (3) 24 (4) 48

65. Volume of one gram equivalent of H_2 at NTP is:-

(1) 5.6 L (2) 22.4 L

(3) 44.8 L (4) 11.2 L

66. How many grams of $AgNO_3$ are needed to prepare 500 mL of a 0.240 M solution ? ($AgNO_3$ = 170 g/mol)

(1) 1.42 g (2) 20.4 g

(3) 120 g (4) 354 g

67. Nitric acid (HNO_3) is 1.6% H, 22.2%N and 76.2% O by mass. All pure samples of HNO_3 have this composition, according to the

(1) Law of multiple proportions

(2) Law of reciprocal proportions

(3) Law of definite proportions

(4) Law of conservation of mass

68. Molarity of 800 g water is

(1) 1 M

(2) Can not be determined

(3) 55.5 M

(4) 0.1 M

69. At same temperature and pressure, calculate the ratio of moles of CO_2 and CH_4 in same volume:

(1) 3 : 5 (2) 2 : 3

(3) 1 : 1 (4) 1 : 2

70. 3.0 molal NaOH solution has a density of 1.110 g/ml. The molarity of the solution is-

(1) 2.9732 (2) 3.05

(3) 3.64 (4) 3.0504

71. Which of the following reactions is not correct according to the law of conservation of mass.

(1) $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

(2) $C_3H_8(g) + O_2(g) \rightarrow CO_2(g) + H_2O(g)$

(3) $P_4(s) + 5O_2(g) \rightarrow P_4O_{10}(s)$

(4) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

72. Vapour density of a gas is 11.2, volume occupied by 2.4g of this at STP will be–

(1) 11.2 L (2) 2.24 L

(3) 22.4 L (4) 2.4 L

73. 4.6×10^{22} atoms of an element weight 13.8 g. The atomic mass of the element is :-

(1) 290 (2) 180

(3) 34.4 (4) 10.4

74. If 1 mL of water contains 20 drops. Then number of molecules in a drop of water is :-

(1) 6.02×10^{23} (2) 1.67×10^{21}

(3) 1.34×10^{18} (4) 4.34×10^{20}

75. One atomic mass unit in kilogram is :-

(1) $\frac{1}{N_A}$ (2) $\frac{1}{1000\,N_A}$

(3) $\frac{12}{N_A}$ (4) $\frac{1000}{N_A}$

76. The number of electrons present in 9.5 g of PO_4^{3-} ions are :-

(1) 4.7 (2) $5\,N_A$

(3) $4.7\,N_A$ (4) $0.1\,N_A$

77. A mixture of gases contain H_2 and O_2 gases in the ratio of 1:16(w/w). What is the molar ratio of the two gases in the mixture ?

(1) 1 : 2 (2) 16 : 1

(3) 1 : 1 (4) 2 : 1

78. $K_2Cr_2O_7 + 14HCl \rightarrow 2KCl + 2CrCl_3 + 3Cl_2 + 7H_2O$

58.8 g $K_2Cr_2O_7$ is reacting with 7.3 g HCl determine volume of chlorine gas at STP ?

(Given Mw of $K_2Cr_2O_7$ = 294)

(1) 1.28 L (2) 2.24 L

(3) 0.96 L (4) 13.4 L

79. The formula of an acid is HXO_2. The mass of 0.0242 moles of the acid is 1.657 g. What is the atomic mass of X ?

(1) 35.5 (2) 28.1

(3) 128 (4) 19.0

80. Number of atoms of nitrogen in 368 amu of N_2O_4 is :-

(1) 8 mol (2) 4

(3) 4 mol (4) 8

81. If 96g of sulphur reacts with 32g of oxygen to form sulphur dioxide. Calculate moles of excess reagent left in the reaction $S + O_2 \longrightarrow SO_2$

(1) 1 mole (2) 3 mole

(3) 2 mole (4) None

82. In a closed vessel 50 ml of A_2B_3 completely reacts with 200 ml of C_2 according to the following equation

$2A_2B_3(g) + 5C_2(g) \longrightarrow 3C_3B_2(g) + CA_4(g)$

The composition of gaseous mixture in the system will be :-

(1) 100 ml C_2, 50 ml C_3B_2, 50 ml CA_4

(2) 25 ml C_2, 75 ml C_3B_2, 25 ml CA_4

(3) 75 ml C_2, 75 ml C_3B_2, 25 ml CA_4

(4) 10 ml C_2, 25 ml C_3B_2, 100 ml CA_4

83. What percentage of oxygen is present in the compound $CaCO_3.3Ca_3(PO_4)_2$? (molecular weight = 1030)

(1) 23.3% (2) 45.36%

(3) 41.94% (4) 17.08%

84. 10 g of a sample of $CaCO_3$ gave 3.3 g of CO_2 on treatment with an excess of HCl. The percentage purity of the sample is :-

$CaCO_3 + 2HCl \longrightarrow CaCl_2 + CO_2 + H_2O$

(1) 50 (2) 60

(3) 70 (4) 75

85. 1.5 g of NaI reacts with 1.6 g of Br_2 to produce 0.412 g of NaBr, then determine the percent yield of the reaction ?

[Atomic weight of Na = 23, Br = 80, I = 127]

$2NaI + Br_2 \longrightarrow 2NaBr + I_2$

(1) 30% (2) 40%

(3) 50% (4) 60%

86. Equal volumes of H_2 and Cl_2 are mixed. How will the volume of the mixture changes after the reaction at same T and P ?

(1) Unchanged (2) Reduced to half

(3) Increases two fold (4) None of these

87. 4.4 g of an unknown gas occupies 2.24 L of volume of NTP. The gas may be:

(1) C_2H_6 (2) CO (3) N_2O (4) CH_4

88. Sum of moles of protons, electrons and neutrons in 16 g of O is :-

(1) 24 (2) 8 (3) 32 (4) 16

89. Which has maximum volume at NTP.

(1) 1.5×10^{23} molecules of CO_2

(2) 1 g H_2

(3) 4 g O_2

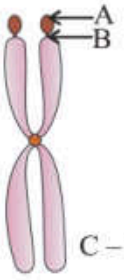
(4) 16 g SO_3

90. Aspirin has the formula $C_9H_8O_4$. How many atoms of oxygen are there in a tablet weighing 360 mg?

(1) 1.204×10^{23} (2) 1.08×10^{22}

(3) 1.204×10^{24} (4) 4.81×10^{21}

91. The accompanying diagram represents a chromosome Identify the structure A, B & type of chromosome C :-



C – Type of chromosome

(1) A-Satellite, B-Primary constriction, C-Acrocentric

(2) A-Satellite, B-Secondary constriction, C-Metacentric

(3) A-Satellite, B-Centromere, C-Telocentric

(4) A-Satellite, B-Centromere, C-Submetacentric

92. **Assertion (A):** Mitochondria are the sites of aerobic respiration.

Reason (R): They produce ATP by the process of photophosphorylation.

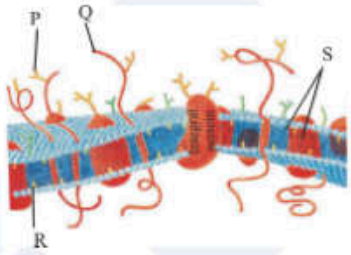
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is incorrect but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**.

93. The figure below show the structure of plasma membrane with its four parts are labelled (P), (Q), (R) and (S). Select the correct match with its function :-



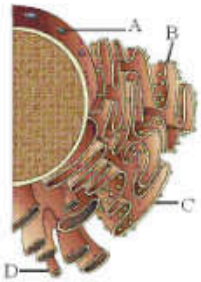
(1) R : Cholesterol : membrane stabilizer

(2) R : Cholesterol : lipid bilayer formation

(3) P : Sugar : cell recognition

(4) (1) & (3) both

94.



Choose the correct option with respect to given diagram

(1) B – Frequently observed in the cells actively involved in protein synthesis and secretion

(2) D – Major site for synthesis of protein–like steroidal hormones in plant cell.

(3) C – Divides the intracellular space into two distinct compartments.

(4) A – Synthesis of r–RNA and ribosome

95. Which of the following is not a function of cell wall ?

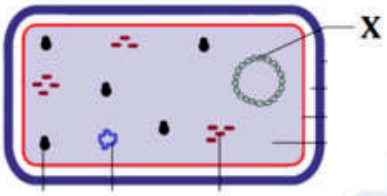
(1) Protection from mechanical damage and infection

(2) Cell to cell interaction

(3) Barrier to undesirable macromolecules

(4) Secretion

96. The given diagram shows a bacterial cell. Identify the part labeled 'X'.



(1) Cell wall (2) Capsule

(3) Plasma membrane (4) Nucleoid

97. An elaborate network of filamentous..... structures present in the cytoplasm is collectively referred to as the ..... respectively:-

(1) Lipoidal, plasma membrane

(2) Cytoskeleton, Proteinaceous

(3) Proteinaceous, Centriole

(4) Proteinaceous, Cytoskeleton

98. Recognise the figure and find out the **correct** statement.



(1) Given figure shows a chromosome with two centromere on surface of it.

(2) Given figure shows a chromosome with two kinetochore on surface of it.

(3) Given figure shows a chromosome with two chromatids attached at primary constriction

(4) Both 2 and 3

99. Read the following statements about cillium or flagellum and find out the **correct** statements.

a. Prokaryotic and eukaryotic flagella are structurally similar.

b. Cilia work like oars and comparatively smaller than flagella.

c. The core of cilium or flagellum is called axoneme.

d. Central tubules are connected by interdoublet bridges and is enclosed by central sheath.

e. Central sheath is connected to one of the tubules of each peripheral doublets by a radial spoke, There are nine radial spokes.

f. Peripheral doublets are also interconnected by A-B linkers [Nexin protein] .

(1) b, d, e, f (2) c, d, e

(3) a, b, c (4) b, c, e, f

100. In which part of cell contains the enzymes for both carbohydrate synthesis and protein synthesis ?

(1) Ribosome

(2) Matrix of mitochondria

(3) Matrix (stroma) of chloroplast

(4) Lysosome

101. Consider the following five statements about prokaryotic and eukaryotic cells and choose the correct option.

I: The genetic material in prokaryotic cells is a circular, naked DNA, not enclosed by a nuclear envelope.

II: The cell wall of bacteria is made up of peptidoglycan.

III: A well-developed cytoskeleton is present in prokaryotic cells.

IV: Mesosomes are infoldings of the cell membrane in prokaryotes that help in DNA replication and cell wall formation.

V: Eukaryotic cells have 80S ribosomes in the cytoplasm and 70S ribosomes in mitochondria and chloroplasts.

(1) Statements I, II, III, and IV are correct.

(2) Statements I, II, IV, and V are correct.

(3) Statements I, III, and V are correct.

(4) All statements are correct.

102. Small bristle like fibres sprouting out of the Bacterial cell, help to attach the bacteria to rocks in streams and also to the host tissues are

(1) Flagella

(2) Mesosomes

(3) Fimbriae

(4) Both cilia and flagella

103. The two sub-units of prokaryotic ribosomes are _____

(1) 50 S and 30 S (2) 80 S and 50 S

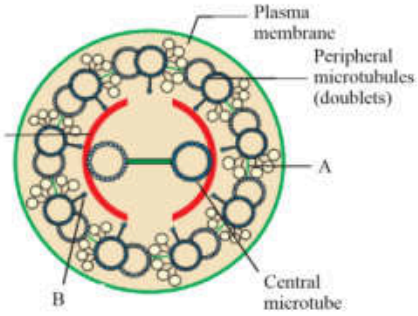
(3) 50 S and 20 S (4) 40 S and 30 S

104. ATP synthesis occur in:

(1) Mitochondria (2) Chloroplast

(3) Both (1) and (2) (4) Leucoplasts

105. Given diagram is of ..... and A and B represents :



(1) Cilia, A-Radial spoke, B-Interdoublet bridge

(2) Flagella, A-Interdoublet bridge, B-Radial spoke

(3) Cilia, A-Central sheath, B-Radial spoke

(4) Centriole, A-Interdoublet bridge, B-Radial spoke

106. The basal body of a cilium is structurally similar to:

(1) Centrosome (2) Centriole

(3) Kinetochore (4) Spindle fibre

107. Which of the following is not true for cilia?

(1) They are hair-like outgrowths of the cell wall.

(2) They cause the movement of either the cell or the surrounding fluid.

(3) Their core is called axoneme.

(4) They are covered with plasma membrane.

108. **Statement-I** : Plasmid DNA confers certain unique phenotypic characters to bacteria.
Statement-II : Plasmid DNA used to monitor bacterial transformation with foreign DNA.

(1) Both Statement I and II correct.

(2) Both Statement I and II incorrect.

(3) Only Statement I correct.

(4) Only statement II correct.

109. Read the following statement and give correct answer regarding bacterial cell
(a) Glycocalyx is the outermost structure in bacterial cell envelope.
(b) The glycocalyx could be a loose sheath called capsule.
(c) The glycocalyx may be the thick and tough called slime layer.

(1) (a) only correct (2) (a), (b) are correct

(3) (a), (c) are correct (4) (b) only correct

110. Read the following statement carefully and identify the correct statements and select the correct option from options given below:
(A) The pili present in prokaryotic cells are involved in active movement of bacteria.
(B) Compartmentalization of cytoplasm is not observed in prokaryotic cell.
(C) Prokaryotic cells are generally smaller and multiply more rapidly than eukaryotic cells.
(D) Gas vacoule is a common character among blue green algae, purple sulphur bacteria and green photosynthetic bacteria.

(1) B and D only (2) A, B and D only

(3) A, B, C and D only (4) B, C and D

111. Consider the two statements:
Assertion (A): In a plant cell, the concentration of a number of ions is significantly higher in the vacuole than in the cytoplasm.
Reason (R): The tonoplast facilitates the transport of a number of ions and other materials against concentration gradients into the vacuole.

(1) Both (A) and (R) are True and (R) is the correct explanation of (A).

(2) Both (A) and (R) are True but (R) is not the correct explanation of (A).

(3) (A) is True but (R) is False.

(4) Both (A) and (R) are False.

112. Given below are two statements:

Statement I: The Golgi cisternae are concentrically arranged near the nucleus with distinct convex cis or the forming face and concave trans or the maturing face.

Statement II: The cis and trans faces of the organelle are identical and interconnected.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are True.

(2) Both Statement I and Statement II are False.

(3) Statement I is True but Statement II is False.

(4) Statement I is False but Statement II is True.

113. Svedberg's unit (S) in ultracentrifugation indirectly provides an analytical measure of an organelle's:

(1) Volume and mass

(2) Density and size

(3) Charge and shape

(4) Length and surface area

114. In chloroplasts, flat membranous tubules that connect the thylakoids of different grana are called:

(1) Golgi apparatus (2) RER

(3) SER (4) Stroma lamellae

115. Read the given statements regarding the plant cell wall and select the correct option:

Statement I: The primary wall of a young plant cell is capable of growth, which gradually diminishes as the cell matures.

Statement II: The secondary wall is formed on the outer side (towards the middle lamella) of the mature plant cell.

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

116. Who proposed the hypothesis that the bodies of animals and plants are composed of cells and products of cells?

(1) Alphonso Corti (2) Matthias Schleiden

(3) Theodore Schwann (4) Rudolf Virchow

117. The detailed structure of the cell membrane was studied only after the advent of electron microscope in the year :-

(1) 1931 s (2) 1913 s

(3) 1950 s (4) 1973 s

118. Membrane bound minute vesicles called microbodies are present:-

(1) In both prokaryotic and eukaryotic cells

(2) Only in cells of fatty seeds

(3) In both plant and animal cells

(4) Only in animal cells

119. Match the column I, II and III and choose the correct combination form the options given:

	Column-I		Column-II		Column-III
a.		1.	Mesophyll cell	K.	Elongated
b.		2.	Tracheid	L.	Round and biconcave
c.		3.	Red blood cells	M.	Amoeboid
d.		4.	White blood cells	N.	Round and oval

(1) a-3-M; b-4-L; c-1-K; d-2-N

(2) a-4-M; b-3-L; c-2-K; d-1-N

(3) a-3-L; b-4-M; c-1-N; d-2-K

(4) a-4-L; b-3-M; c-2-N; d-2-K

120. In the given four statements (a-d) select the options which includes all the correct ones :-

(a) Simple epithelium is composed of single layer of cells and functions as a lining for body cavities, ducts and tubes

(b) The squamous epithelium is made of single thin layer of flattened cells with regular boundries

(c) Microvilli are found in the lining of stomach and intestine and help in secretion and absorption

(d) The function of cilia is to move particles or mucus in a specific direction over the epithelium

(1) a, b, c (2) b, c, d

(3) a, c, d (4) a, d

121. Out of following which structure does not have non-keratinized stratified squamous epithelium :

(1) Pharynx (2) Oesophagus

(3) Cornea of eye (4) Epidermis

122. Identify the correct match from the column I, II and III :-

	Column - I	Column - II		Column - III	
(1)	Bones	(a)	Adipocytes	(i)	Solid matrix
(2)	Cartilage	(b)	Fibroblast cells	(ii)	Loose matrix
(3)	Areolar tissue	(c)	Osteocytes	(iii)	Store fat
(4)	Adipose tissue	(d)	Chondrocytes	(iv)	Hard matrix

(1) (1)-(d)-(i); (2)-(c)-(iv); (3)-(a) (iii); (4)-(b)-(ii)

(2) (1)-(c)-(iv); (2)-(d)-(i); (3)-(b)-(ii); (4)-(a)-(iii)

(3) (1)-(c)-(i); (2)-(d)-(iv); (3)-(b)-(iii); (4)-(a)-(ii)

(4) (1)-(a)-(i); (2)-(b)-(ii); (3)-(c)-(iii); (4)-(d)-(iv)

123. Urothelium tissue found in :-

(1) Brain ventricles (2) Fallopian tubes

(3) Urinary bladder (4) Bronchioles

124. Which of the following epithelium is present in the inner lining of ducts of salivary glands and pancreatic ducts ?

(1) Simple epithelium

(2) Stratified cuboidal epithelium

(3) Columnar epithelium

(4) Stratified columnar epithelium

125. How many of the following substances are secreted by exocrine glands.

mucus, thyroxine, saliva, earwax, insulin, oil, milk & digestive enzymes:-

(1) Four (2) Five (3) Six (4) Seven

126. Identify wrong match :

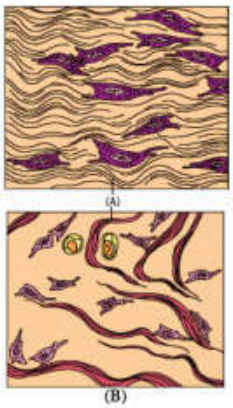
(1) Tight junction - Allow substance from leaking across a tissue

(2) Endocrine gland - duct is absent

(3) Glandular epithelium cells - Cuboidal and Columnar

(4) Simple squamous epithelium - help in diffusion

127. Select the correct answer related to given diagram A and B -



(1) 'A' is found in both Tendon and Ligament

(2) 'B' is found in both Tendon and Ligament

(3) 'A' is found in Tendon while 'B' is found in Ligament

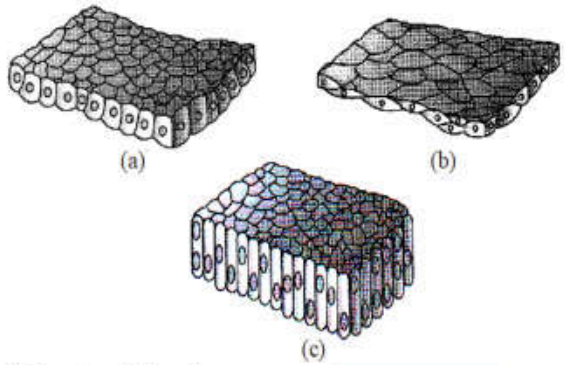
(4) 'B' is found in Tendon while 'A' is found in Ligament

128. Which property is provided by fibres for tissue?

(1) Strength (2) Elasticity

(3) Flexibility (4) All

129. Given below are diagrammatic sketch of some types of simple epithelium. Identify them and match with their suitable locations :



(i) Ducts of glands
(ii) Lining of intestine
(iii) Dry surface of skin
(iv) Walls of blood vessels
(v) PCT
(vi) Air sacs of lungs

(1) a-v & iii, b-iv & ii, c-vi

(2) a-iv & vi, b-i & ii, c-iii

(3) a-i & v, b-iv & vi, c-ii

(4) a-iv, b-ii & v, c-i & iii

130. Choose the correct option:

Assertion: Epithelial tissues have a free surface facing either a body fluid or the outside environment.

Reason: This orientation is crucial for their function in covering or lining parts of the body.

(1) Both Assertion and Reason are correct, and Reason is the correct explanation for Assertion

(2) Both Assertion and Reason are correct, but Reason is not the correct explanation for Assertion

(3) Assertion is correct, but Reason is incorrect

(4) Assertion is incorrect, but Reason is correct

131. ***Assertion***: Tight junction facilitate the cell to communicate with each other by connecting the cytoplasm.

Reason: Gap junction help to stop substances from leaking across a tissue.

(1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

132. Match the following simple epithelial tissues in column I with their occurrence in column II and choose the correct combination from the options given

Column I		Column II	
A	Squamous	1	Intestinal glands
B	Cuboidal	2	Trachea
C	Columnar	3	Ovary
D	Ciliated	4	Blood vessels
E	Pseudo stratified	5	Bronchioles

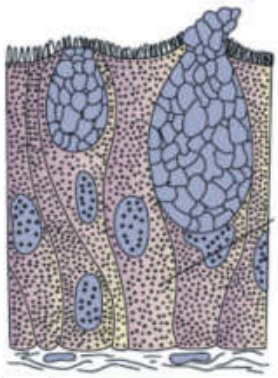
(1) A – 5, B – 4, C – 2, D – 1, E – 3

(2) A – 4, B – 5, C – 1, D – 2, E – 3

(3) A – 4, B – 3, C – 1, D – 2, E – 5

(4) A – 4, B – 3, C – 1, D – 5, E – 2

133. Identify the given gland below as well as the related secretion from it and select the right option for the two together.



	Gland	Product
(1)	Multicellular gland	Mucus
(2)	Unicellular gland	Ear wax
(3)	Sweat gland	Sweat
(4)	Goblet cell	Mucus

134. Read the following five statements [A–E] :-

(A) It forms the lining of the cavities of alveoli of lungs

(B) It forms the lining of wet surface like buccal cavity and oesophagus

(C) It occurs in the ducts of glands

(D) It forms the inner lining of duct of salivary glands and pancreatic duct

(E) It is a loose connective tissue

Which of the above are associated with simple epithelial tissue ?

(1) A and C (2) B and C

(3) A and D (4) D and E

135. **Assertion:** Squamous epithelium helps in the diffusion of gases in lungs.

Reason: Squamous epithelium bears microvilli.

(1) Both assertion and reason are true and reason is correct explanation of assertion

(2) Both assertion and reason are true, reason is not the correct explanation of assertion

(3) Assertion is true, reason is false

(4) Both assertion and reason are false

136. Basement membrane is:-

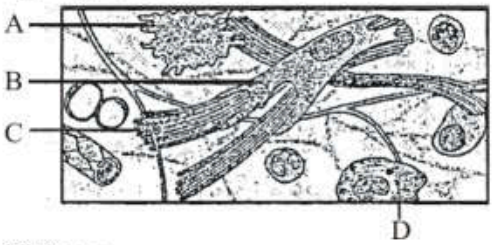
(1) cellular, non-living

(2) non-cellular, living

(3) cellular, living

(4) non-cellular, non-living

137. Given below is the diagrammatic representation of a certain type of connective tissue. Identify the parts labelled A, B, C and D, and select the right option about them.



Options :-

	Part-A	Part-B	Part-C	Part-D
(1)	Macro-phage	Collagen fibres	Fibroblast	Mast cell
(2)	Mast cell	Collagen fibres	Fibroblast	Macro-phage
(3)	Macro-phage	Fibroblast	Collagen fibres	Mast cell
(4)	Mast cell	Macro-phage	Fibroblast	Collagen fibres

138. Salivary glands are :-

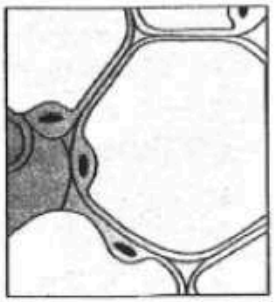
(1) Unicellular – exocrine gland

(2) Multicellular – exocrine gland

(3) Multicellular - endocrine glands

(4) Specialised ciliated epithelium

139. Following is a diagram of connective tissue. Find the odd one out w.r.t its location



(1) Beneath skin (2) Around kidney

(3) Around eyeball (4) Cartilage

140. **Assertion (A) :-** Most of the bones are cartilaginous in origin.

Reason (R) :- Most of the embryonic cartilage goes under the process of ossification.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

141. The matrix of connective tissue is of :

(1) Lipids

(2) Monosaccharides

(3) Modified polysaccharide

(4) Phospholipids

142. Simple cuboidal epithelium in gonads is also called

(1) Pavement epithelium

(2) Germinal epithelium

(3) Urothelium

(4) Tessalated epithelium

143. Mammalian bone differs from bones of other animals due to presence of :-

(1) Collagen (2) Blood vessels

(3) Lymph vessels (4) Haversian canal

144. Intervertebral disc and pubis symphysis represent which cartilage:

(1) Calcified (2) Elastic

(3) Hyaline (4) White fibrous

145. Which of the following type of epithelium is present in trachea?

(1) Simple columnar

(2) Pseudo-stratified columnar

(3) Stratified columnar

(4) Transitional

146. Cilia, which are contractile process occurs in, except

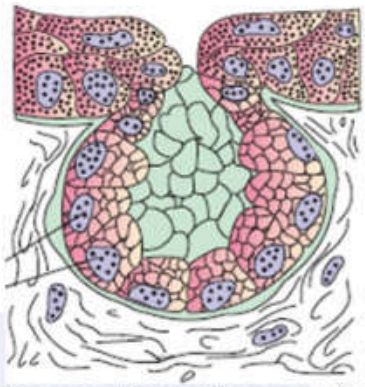
(1) Pseudostratified epithelium of trachea

(2) Columnar epithelium of bronchioles

(3) Columnar epithelium of stomach

(4) Columnar epithelium of oviduct

147.



Choose the appropriate option w.r.t given figure.

(1) Goblet cell (2) Salivary gland

(3) Mammary gland (4) Both (2) & (3)

148. Basement membrane is secreted by :-

(1) Connective tissue (2) Epithelial tissue

(3) Muscular tissue (4) Both 1 & 2

149. A heart is considered a complex organ because it:

(1) Contains only muscular tissue

(2) Is made only of connective and epithelial tissues

(3) Contains all four basic types of tissues

(4) Contains only neural tissue for regulation

150. **Statement-I:** In human erythrocyte about 50 percent lipid and 40 percent protein are present.
Statement-II: Protein show lateral movement in phospholipid bilayer due to quasi fluid nature of lipid.

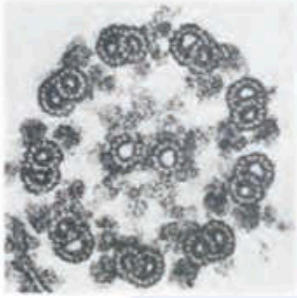
(1) Both statement-I and II are correct.

(2) Both statement-I and II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

151. Identify the diagram and find the option with correct arrangement present in the organelle.



(1) Centriole- 9+0 arrangement

(2) Cilia- 9+0 arrangement

(3) Flagella- 9+2 arrangement

(4) Centriole- 9+2 arrangement

152. **Statement-I :** Nucleolus is not a membrane bound structure.
Statement-II : Nucleolus is the site for active messenger RNA synthesis.

(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect.

(3) Only Statement-I is correct.

(4) Only Statement-II is correct.

153. Middle lamella mainly consists of :-

(1) Ca-pectate (2) Mn-pectate

(3) K-pectate (4) Na-pectate

154. Match the following columns & select the correct option :

Column-I		Column-II	
(a)	RER	(i)	Steroidal hormones synthesis
(b)	SER	(ii)	Protein synthesis
(c)	Primary cell wall	(iii)	Cytoplasmic connections
(d)	Plasmodesmata	(iv)	Capable of growth

(1) a-ii, b-iii, c-iv, d-i (2) a-ii, b-i, c-iii, d-iv

(3) a-ii, b-i, c-iv, d-iii (4) a-i, b-ii, c-iv, d-iii

155. Which of the following are considered as the part of endomembrane system ?

(A) Mitochondria (B) Endoplasmic Reticulum

(C) Chloroplast (D) Golgi complex

(E) Lysosome (F) Peroxisome

Choose the most appropriate answer from the options given below :-

(1) A, C & E only (2) B, D & E only

(3) C, D & F only (4) A & B only

156. Which of the following cell organelle is composed of ribonucleic acid and protein and are not surrounded by any membrane ?

(1) Golgi body

(2) Endoplasmic reticulum

(3) Ribosome

(4) Mitochondria

157. Which of the following structures are not present in the structure of a chromosome ?

A. Chromatid B. Centrosome

C. Satellite D. Kinetochore

E. Axoneme

(1) A, B, C & D (2) A, C & D

(3) B & E (4) Only E

158. Which of the following is incorrect with reference to membrane transport?

(1) Neutral solutes can move across the membrane by simple diffusion along the concentration gradient

(2) Movement of water across the membrane along the concentration gradient is known as osmosis

(3) Polar molecule requires carrier protein of the membrane for transport across the membrane

(4) ATP is utilized in facilitated diffusion of polar uncharged molecules along the concentration gradient

159. Which of the following statements are correct?

(A) Virchow modified cell theory in 1838.

(B) Unicellular organisms are capable of performing the essential functions of life.

(C) Anything less than a cell ensure independent living.

(D) Some organisms are composed of single cell and are called unicellular organisms.

(1) A, B and D only (2) B and D only

(3) A and C only (4) A, B, C and D all

160. Which one of the following cellular part is correctly described ?

(i) Ribosomes in mitochondria are 80-S type

(ii) Lysosomes optimally active at the alkaline pH.

(iii)Centrioles are the sites for RNA synthesis.

(iv)Thylakoids have system of ATP formation.

(v) Smooth endoplasmic reticulum is the major site for synthesis of lipids.

(1) (i), (ii) and (iii) (2) (iii), (iv) and (v)

(3) (iv) and (v) (4) (i), (ii) and (v)

161. Read the following statement carefully :-

(A) Nucleolus as a cell organelle was first described by Robert Brown in 1831.

(B) The nucleoplasm contain nucleus & chromatin

(C) The nuclear envelope space do not act as barrier between materials present inside nucleus and cytoplasm.

(D) Chromatin contain DNA, RNA, histone & non-histone protein

Choose the correct option which contain correct statement :-

(1) A, B & C (2) B & D

(3) Only D (4) B, C & D

162. During the different stages of cell division, cells show chromosomes in place of the nucleus. The chromosomes can be classified into four types,such as metacentric, submetacentric, acrocentric and telocentric.What is the basis of this categorization of chromosomes ?

(1) Length of chromosomes

(2) Position of the secondary constriction

(3) Length of DNA

(4) Position of the centromere

163.



Consider the given figure and which of the following are not concern with it ?

(a) Divide by fission

(b) Produce cellular energy (ATP)

(c) Chlorophylls are found

(d) Matrix possesses enzymes for dark reaction

(e) Site of aerobic respiration

(f) Ribosomes are smaller than cytoplasmic ribosome

(1) a, b, e, f (2) a, b, c, d

(3) c, d, e, f (4) c, d

164. Match the columns I and II, and choose the correct combination from the options given

	Column I (organelle)		Column II (Size)
a.	Cisternae of GB	K.	2–4μm
b.	Length of mitochondria	L.	0.2–1.0μm
c.	Diameter of mitochondria	M.	5–10μm
d.	Length of chloroplast	N.	0.5–1.0μm
e.	Width of chloroplast	Q	1.0–4.1μm

(1) a–L, b–M, c–N, d–Q, e–K

(2) a–N, b–Q, c–K, d–M, e–L

(3) a–L, b–Q, c–N, d–M, e–K

(4) a–N, b–Q, c–L, d–M, e–K

165. Arrange the following in decreasing order of their typical size:-

(A) Bacteria (B) Virus

(C) Mycoplasma (D) RBC

(1) D > A > C > B (2) D < A < C < B

(3) B < A < C < D (4) A > D > C > B

166. Match column I with column II and select the correct option.

Column I		Column II	
A.	Ciliated epithelium	(i)	Secrete enzyme, hormone and other substances
B.	Compound epithelium	(ii)	Form diffusion boundary
C.	Squamous epithelium	(iii)	Protection against chemical stress
D.	Glandular epithelium	(iv)	Movement of mucus in one direction

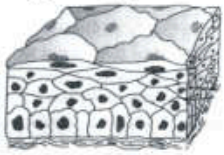
(1) A – (iii), B – (ii), C – (iv), D – (i)

(2) A – (i), B – (iv), C – (ii), D – (iii)

(3) A – (ii), B – (iii), C – (i), D – (iv)

(4) A – (iv), B – (iii), C – (ii), D – (i)

167. Read the following statements and find out how many of these are related to given figure :-



a. Multilayered epithelium

b. Limited role in secretion and absorption

c. Main function is to provide protection against chemical and mechanical stresses.

d. They cover the dry surface of skin, moist surface of buccal cavity and pharynx.

(1) Four (2) Three

(3) Two (4) One

168. Bone have a hard and non-pliable ground substance rich in ___(A)___ salts and ___(B)___ fibres :-

(1) (A) – Magnesium, (B) – Elastin

(2) (A) – Calcium, (B) – Collagen

(3) (A) – Phosphorous, (B) – Reticular

(4) (A) – Calcium, (B) – Elastin

169. Simple cuboidal epithelium is found in :-

(1) Follicle of thyroid gland

(2) Tubular part of nephron

(3) Both 1 and 2

(4) Brain

170. Which type of epithelium have elongated cells with basal nuclei :-

(1) Epithelium of PCT

(2) Epithelium of air sac

(3) Epithelium of stomach

(4) Epithelium of blood vessels

171. Which one of the following types of cell is involved in making of the inner walls of large blood vessels.

(1) Cuboidal Epithelium

(2) Columnar Epithelium

(3) Squamous Epithelium

(4) Stratified Epithelium

172. Lining of coelom is :-

(1) Endothelium (2) Mesothelium

(3) Endocardium (4) Endomycium

173. How many statements are true regarding "Epithelial tissue" ?

(a) Epithelial cells are closely packed with each other so there is very little intercellular space.

(b) During embryonic development epithelium originates after connective tissue.

(c) It has high regeneration power.

(d) For the nutrients it depends on underlaying connective tissue.

(1) One (2) Two

(3) Three (4) Four

174. ***Assertion :-*** Products of endocrine glands are called hormones are secreted directly into the fluid bathing the gland .

Reason :- Endocrine glands do not have ducts.

(1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

175. All the following statements are correct about bones, but one is wrong. Which one is wrong?

(1) Bones have a hard and non-pliable ground susbtance rich in calcium salts and collagen fibres which give bone its strength

(2) Red bone marrow present in diaphysis is the site of production of blood cells

(3) One osteocyte is present in one lacuna

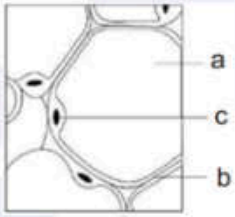
(4) Haversian canals contain nerves and blood vessels

176. How many cartilages in given list is an example of elastic cartilage ?

Pubis symphysis, Pubis of frog, Head of femur, Tip of nose, Epiglottis, Nasal septum, Articular cartilage, C-shaped rings of trachea, Wall of Eustachian tube

(1) Seven (2) Five (3) Three (4) Two

177. Identify a, b and c in the given figure.



(1) a - nucleus, b - fat storage area, c - plasma membrane

(2) a - fat storage area, b - nucleus, c - plasma membrane

(3) a - fat storage area, b - cell wall, c - nucleus

(4) a - fat storage area, b - plasma membrane, c - nucleus

178. Bone destroying cells are called :

(1) Osteocytes (2) Osteoblast

(3) Osteoclast (4) Chondroclast

179. Find out the odd one on basis of number of cells:

(1) Salivary gland (2) Goblet cell

(3) Pituitary gland (4) Thyroid gland

180. Which tissue serves as a support framework for epithelium :

(1) Adipose C.T.

(2) Areolar C.T.

(3) Dense regular C.T.

(4) Dense irregular C.T.

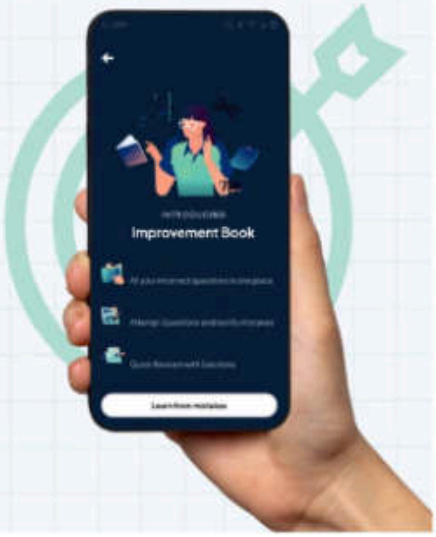
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